

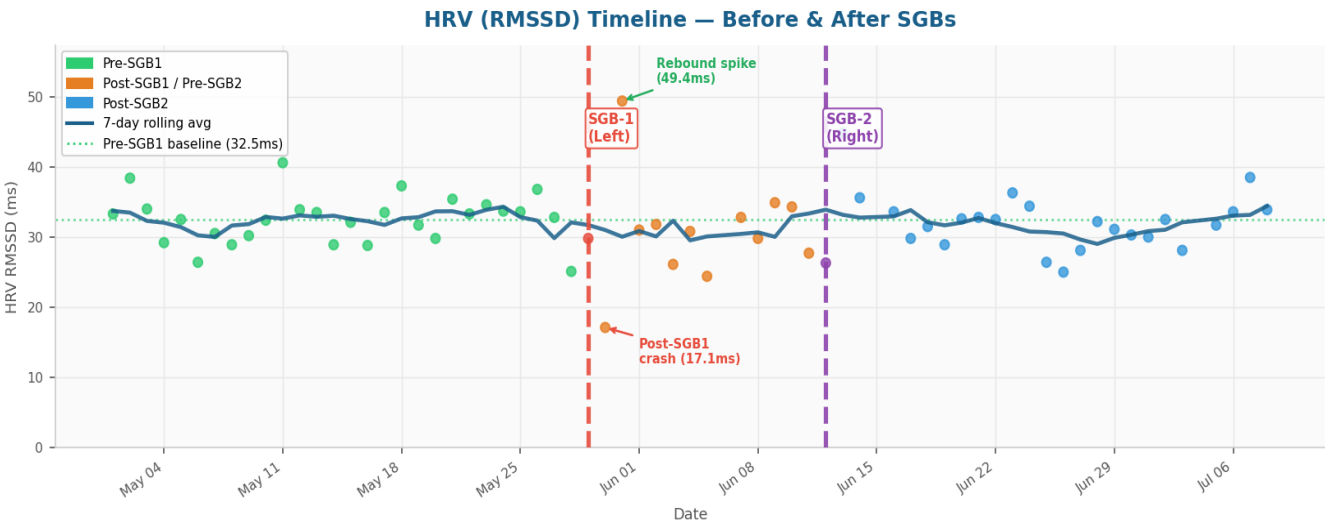
Stellate Ganglion Block (SGB) Response Report

Hannah Munguia — Biometric & Symptom Analysis
Data: WHOOP + Cadence Self-Report | Period: May 1 – July 8, 2026

Procedure Summary

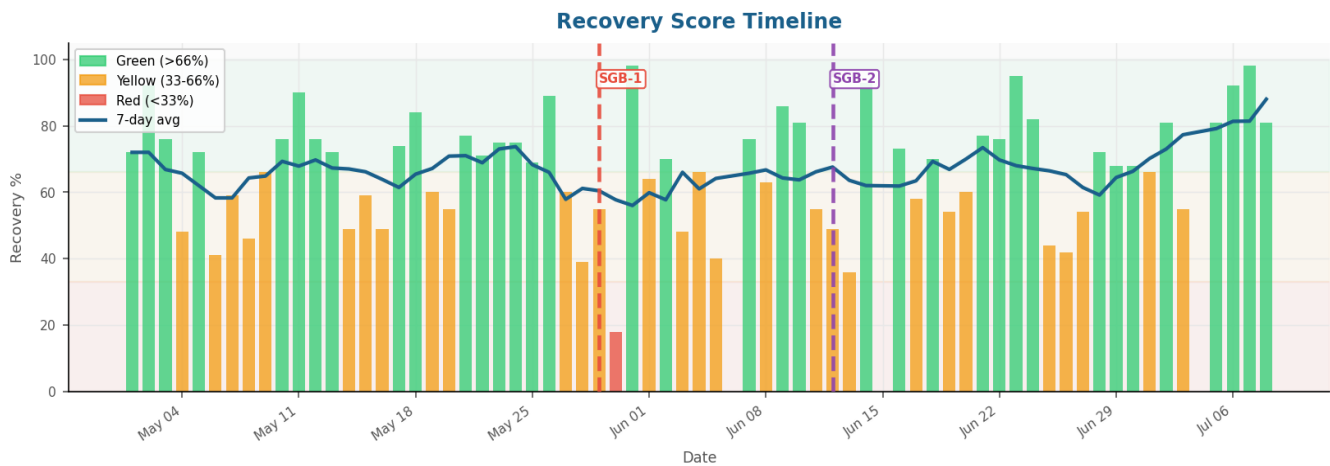
	SGB-1 (Left Side)	SGB-2 (Right Side)
Date	Friday, May 29, 2026	Saturday, June 12, 2026
Side	Left	Right
Day Type	High-demand (drive + wait + procedure)	Moderate (procedure + evening sewing)
Pre-procedure HRV	29.8 ms	26.3 ms
Pre-procedure Recovery	55%	49%
Nadir (lowest day)	May 30: HRV 17.1ms, Recovery 18%	Jun 13: sleep 7%, no overnight sleep
Stabilization	Jun 2 (D+4): HRV 31.8, Recovery 70%	Jun 16 (D+4): HRV 33.6, Recovery 73%

HRV (RMSSD) — Primary Outcome



Key observations: Pre-SGB1 HRV baseline averaged **32.5 ms** (May 1–28). SGB-1 caused an acute crash (17.1 ms, May 30) followed by an extraordinary rebound spike (49.4 ms, May 31). David noted this was the fastest HRV turnaround ever observed in Hannah. Post-SGB2, HRV has been more consistently in the 30–38 ms range with recent readings (Jul 6–7) reaching 33.6 and 38.5 ms. The 7-day rolling average shows a modest upward trend in the post-SGB2 period.

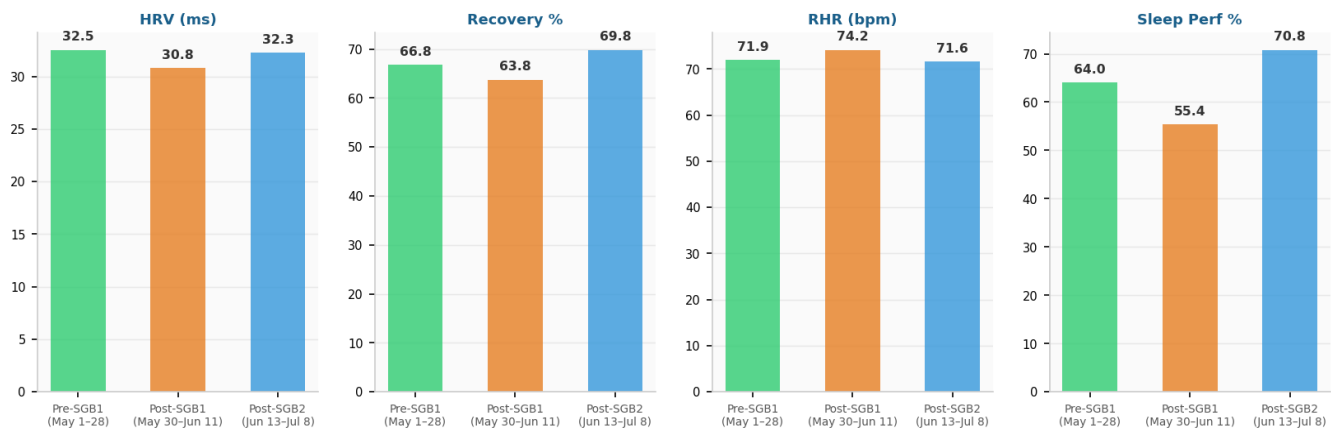
Recovery Score



Key observations: Recovery crashed to 18% the day after SGB-1 then spiked to 98% (May 31). SGB-2 was better tolerated — recovery stayed in yellow/green range the following days. Recent weeks (Jun 20–Jul 8) show more consistent green-zone recovery scores.

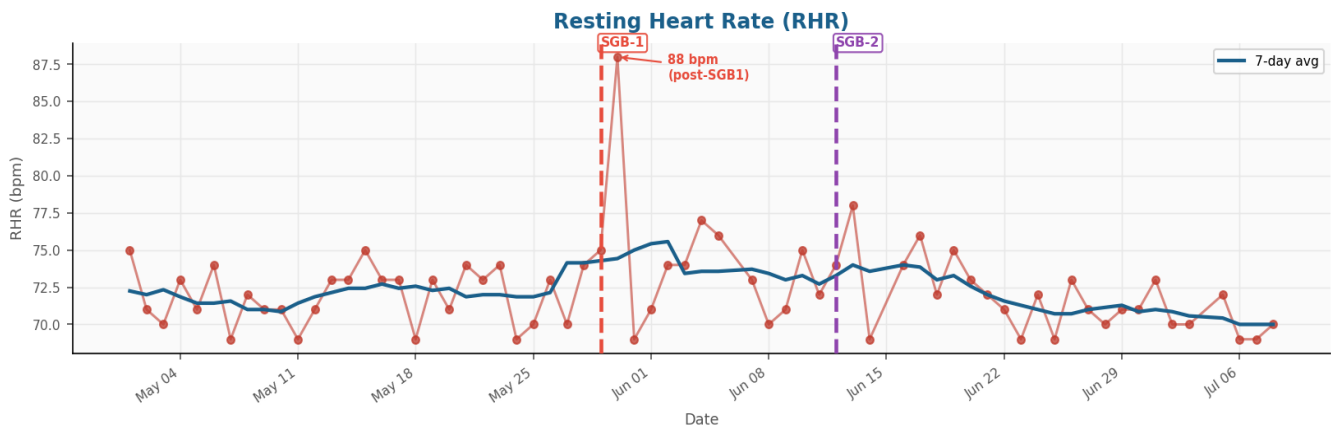
Period Averages: Pre vs. Post SGB

Before vs. After: Average Biometrics by Period



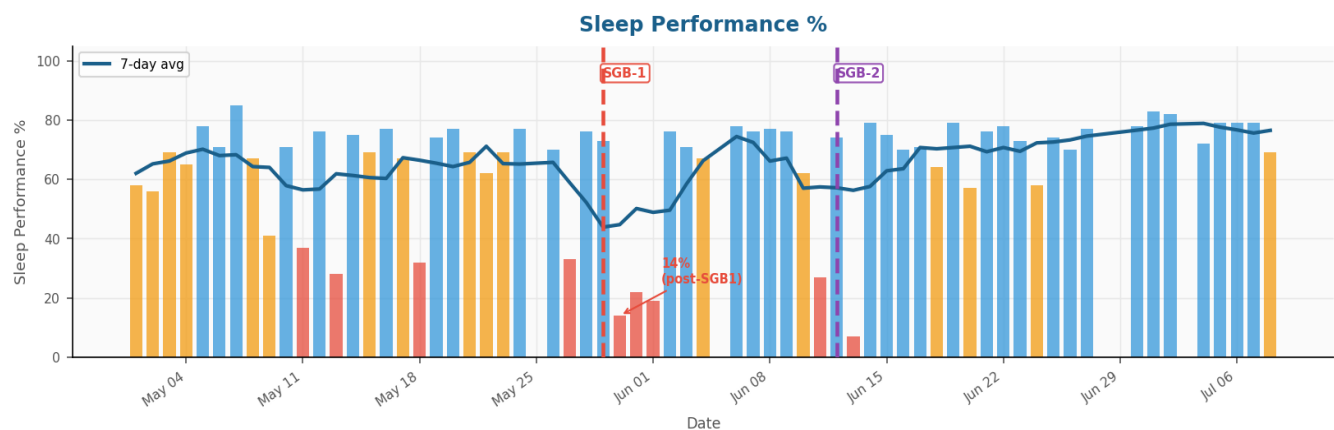
Period	HRV (mean±SD)	Recovery (avg)	RHR (avg)	Sleep Perf (avg)
Pre-SGB1 (May 1–28)	32.5±3.4	67%	72	64%
Post-SGB1 (May 30–Jun 11)	30.8±7.3	64%	74	55%
Post-SGB2 (Jun 13–Jul 8)	32.3±4.0	70%	72	71%

Resting Heart Rate (RHR)



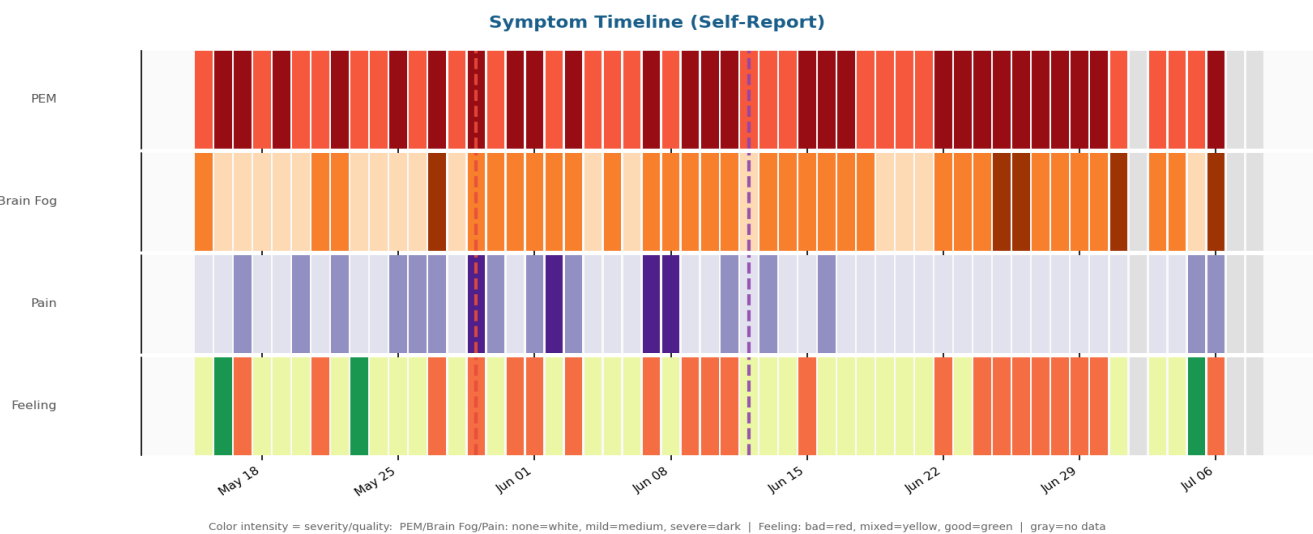
Key observations: RHR spiked to 88 bpm the day after SGB-1 (sympathetic rebound/stress response). Both pre-SGB baselines and post-SGB2 settled values are consistently in the 69–75 bpm range. No sustained elevation post-SGB2 — better procedure tolerance.

Sleep Performance



Key observations: Sleep performance crashed to 14% the night after SGB-1 (couldn't sleep, 2am awake). Sleep then disrupted for ~3 nights (night-time sleep absent, only naps). Post-SGB2, sleep scores have been notably more stable with several 78–83% nights. Recent week (Jul 5–8) shows consistent 69–82% sleep performance — an improvement over pre-SGB levels.

Symptom Timeline (Self-Report)



Each column = one day. Color intensity reflects severity. PEM/Brain Fog/Pain: white = none, medium = mild, dark = severe. Feeling: green = good, yellow = mixed, red = bad. Gray = no data.

SGB-1 (Left Side) — Symptom Log

May 29, 2026

Day	Date	Feeling	PEM	Brain Fog	Pain	Key Symptoms / Notes
SGB Day	May 29	Bad	Severe	Mild	Severe	Hoarse voice, neck pain, emotional discharge, Horner's sign (eye droop), unable to nap, 'wired but tired'
D+1	May 30	Mixed	Mild	Mild	Mild	Woke 4:30am; hoarseness resolved; hip/body aches, bloating, constipation, flushed cheeks. HRV 1
D+2	May 31	Bad	Severe	Mild	None	Increased smell sensitivity (olfactory acuity). Insomnia at 2:20am — strange headache at back of ne
D+3	Jun 1	Bad	Severe	Mild	Mild	Neck feels bruised at injection site. Severely fatigued — hard time moving (alarming level of fatigue)
D+4	Jun 2	Mixed	Mild	Mild	Severe	First time outside (wheelchair). Intense uterine cramping ~8pm. Very sore knees. HRV 31.8 (stabiliz
D+5	Jun 3	Bad	Severe	Mild	Mild	Home all day. Tried to sew — couldn't
D+6	Jun 4	Mixed	Mild	None	None	Some sewing, light housework. Signs of stabilization
D+7	Jun 5	Mixed	Mild	Mild	None	Washed sewing fabric. Relatively good day

Note: SGB-1 was a high-demand day (driving to/from clinic, wait room, procedure). The crash severity may be partly attributable to the demand level, not the procedure alone.

SGB-2 (Right Side) — Symptom Log

June 12, 2026

Day	Date	Feeling	PEM	Brain Fog	Pain	Key Symptoms / Notes
SGB Day	Jun 12	Mixed	Mild	None	None	Much better tolerated than SGB-1. Minor eye droop only — no hoarseness, no dissociation, normal
D+1	Jun 13	Mixed	Mild	Mild	Mild	No sleep overnight. Slight headache (back of head), flushed cheeks. Trip to Newport with Harrison
D+2	Jun 14	Mixed	Mild	Mild	None	Beach time, sun, standing. Sudden fatigue → long nap. 2hr car ride home.

D+3	Jun 15	Bad	Severe	Mild	None	Xolair injection (same day). Demanding day — confounding with Xolair response.
D+4	Jun 16	Mixed	Severe	Mild	Mild	Recovering
D+5	Jun 17	Mixed	Severe	Mild	None	
D+6	Jun 18	Mixed	Mild	Mild	None	Claunch appointment
D+7	Jun 19	Mixed	Mild	None	None	Signs of stabilization

Note: The immediate post-procedure experience was significantly better than SGB-1. The subsequent crash (Jun 15+) is confounded by a Xolair injection and Newport travel on D+1/D+2.

Clinical Observations & Context

Observation	Detail
SGB-1 vs SGB-2 Tolerance	SGB-2 was dramatically better tolerated. Hannah experienced minimal side effects (minor eye droop only, no hoarseness, no dissociation).
Autonomic Rebound (SGB-1)	The HRV spike to 49.4 ms on May 31 (D+2 post-SGB1) was described by David as the fastest HRV turnaround ever observed in Hannah.
Olfactory Acuity	Hannah reported heightened smell sensitivity on D+2 (May 31 — 'super sensitive to the litter box'). This is a recognized early SGB2 sign.
Horner's Sign (SGB-1 only)	Eye droop on the day of SGB-1 confirmed the stellate ganglion was blocked. Nurse noted 'strong autonomic response'. This was a good sign.
Confounders	Several factors complicate clean before/after analysis: (1) LDN dose changes in May (6mg→4.5mg→3mg), (2) Prozac changes (10mg→20mg), (3) Propranolol added.
Post-SGB2 Trend (Late June)	A prolonged low-feeling period (late Jun 22–30) followed SGB-2 with multiple bad/severe PEM days. Several external stressors noted.
Recent Trajectory (July)	July 4–8 shows encouraging signs: Hannah sewed on Jul 4 (first time since before Father's Day), reported 'good' feeling on Jul 5, and HRV improved.
Nicotine Patch Trial (Jul 6)	Hannah tried a half nicotine patch (3.5 mg) which caused severe fatigue within 1 hour, unable to move. Removed at hour 4. This suggests nicotine sensitivity.

Active Medications During Study Period

Medication	Notes
LDN (Low-Dose Naltrexone)	Started at 6mg; reduced to 4.5mg (May 23), then 3mg (May 27) around SGB-1 period
Prozac (fluoxetine)	Changed from 20mg → 10mg (May 26), then taper to 0 (Jun 11+)
Propranolol	Taken before SGB-1. 20mg AM added Jun 12.
Xolair (omalizumab)	Injection Jun 15 — confounds early post-SGB2 period
Rapamycin	6mg single dose May 26 (day before SGB-1 crash)
BCP-157 (oral peptide)	Started May 21
Pepzin / Pepzid (famotidine)	Taken intermittently for GI/MCAS support
VNS device (Pulsetto)	Not used night of SGB-1 (per provider instruction). Resumed after.